

Machine Learning Basics Assignment

This sample assignment demonstrates the basics of Machine Learning using a simple classification model. The assignment includes dataset preprocessing, model training, testing, and evaluation.

1. Dataset

The Iris Dataset is used for classification.
Features: Sepal Length Sepal Width Petal Length Petal Width Target: Flower Species

2. Algorithm Used

Logistic Regression is used as the Machine Learning algorithm. The model is trained using Scikit-learn.

3. Sample Python Code

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = LogisticRegression()
model.fit(X_train, y_train)
accuracy = model.score(X_test, y_test)
print(accuracy)
```

4. Results

Metric	Value
Accuracy	96%
Dataset	Iris Dataset
Algorithm	Logistic Regression

5. Conclusion

This assignment demonstrates the basic workflow of a Machine Learning classification project using

Python and Scikit-learn.